Otopalik Lab Print Chamber Fabrication Procedure

Last Edited: 05/13/2024 – Erin Solomon

Many fly behavior labs at Janelia use backlight illumination for visualization of fly behavior. In these experimental setups, the bottom surface of the behavioral arenas transmits visible light for optogenetic activation and infrared (IR) light to generate sufficient contrast for videos recorded by overhead cameras. The Otopalik lab requested an adaptation of this standard behavior system that incorporates an overhead projector. In this system, a dark fly-sized circle or ellipse is projected onto the walking surface of the arena from above. The real fly may then interact with this projected stimulus. This feature is loosely based on that of a system designed in Vanessa Ruta's lab at The Rockefeller University. To incorporate overhead projection into the optogenetic behavior rig, I designed a series of 3D printed parts with enough opacity for display of the projected stimulus, yet sufficient transmission of visible and IR light for optogenetic activation and overhead imaging of the flies.

The parts printed for these assays are the PFly Bowl Single_Multi Material (Fig. 1), the PFly Bowl_90MM (Fig. 2), and the Universal Bubble Array Diffusor Print (Fig. 3). These parts are multi-material parts – meaning that they are printed as one solid body, but contain features made from different materials. They were designed for different behavioral assays with various geometries specific to each experiment. However, the throughline connecting feature is that they are designed with two main layers – a top white layer, on which the projected stimuli are visible and with which the flies appear in high contrast on video recordings; and a bottom clear layer, which increases structural stability without affecting the amount of backlight illumination that passes through the printed part. The Universal Bubble Array Diffusor Print also has three black circles, to demarcate the three chamber Regions of Interest in the videos that will later be processed by the Branson lab pipeline.

The multi-material printing is achieved using a Stratasys J850 multi-jet printer. The white, clear, and black materials used in these prints are Vero White, Vero Clear, and Vero Black. There are several key advantages to manufacturing these parts in this way. First, by using these parts the researcher can successfully include both backlight optogenetic activation and overhead projected visual stimuli in their experiments. Second, because of the way these parts are designed in CAD the thickness and relative position of the white and clear layers are known and constant quantities, regardless of the overall geometry of the part; so the intensity of the light for optogenetic activation can be measured and kept relatively similar across behavioral assays and printed parts.

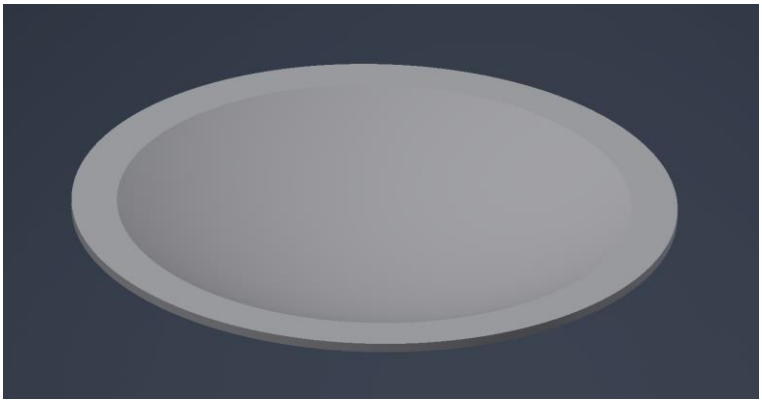


Figure 1.

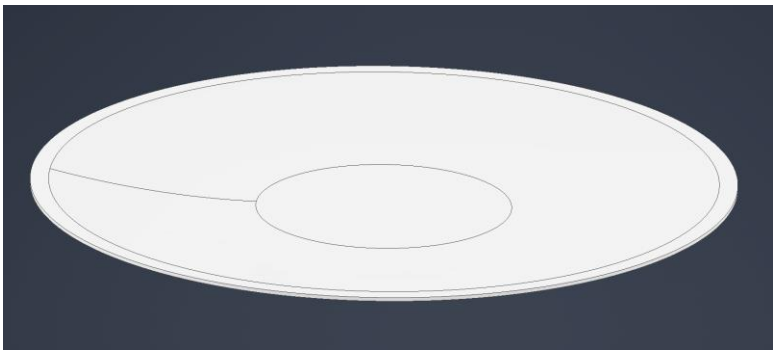


Figure 2.

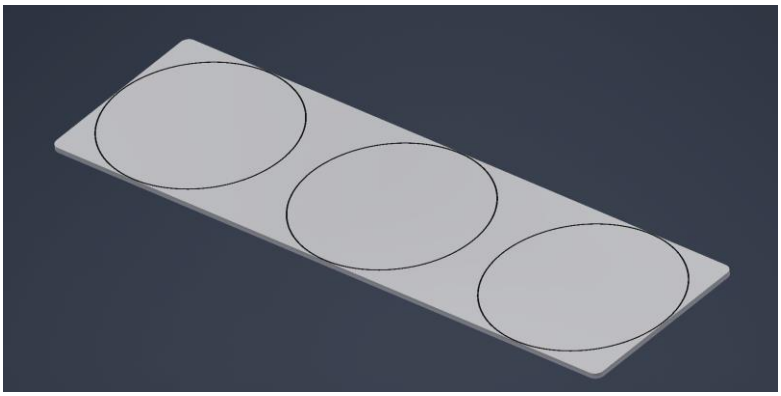


Figure 3.

The CAD models were made in Autodesk Inventor, as .ipt part files. They are built such that each portion of the part that will be printed in a specific color is an independent solid body. The bodies can be found in the Solid Bodies tab underneath the part name on the lefthand browser tree of Inventor (Fig. 4). Within this folder, all bodies have been renamed to indicate the print color and which portion of the overall part they comprise (Fig. 5).

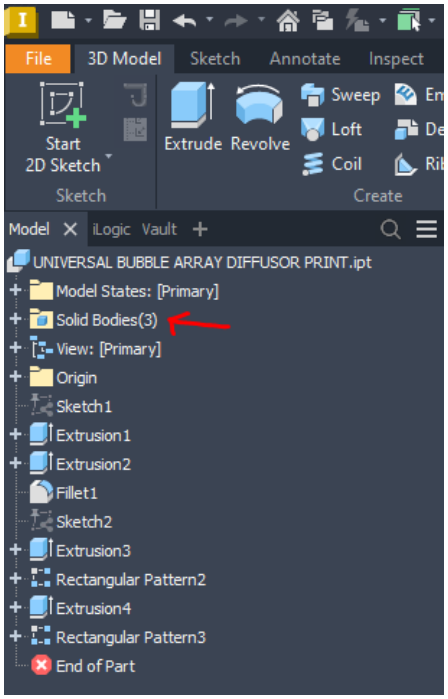


Figure 4.

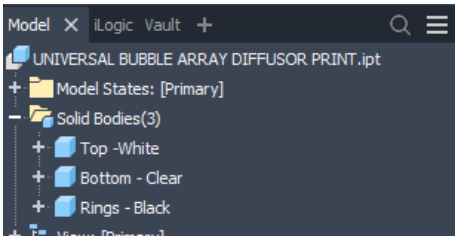


Figure 5.

Each one of these bodies needs to be exported as an individual .stl file. To do this, right click on the first solid body and select “hide others” from the dropdown menu (Fig. 6). Using the Universal Bubble Array Diffusor Print as an example, the black rings and clear bottom half of the part should now be hidden. (Compare Figs. 7 and 8).

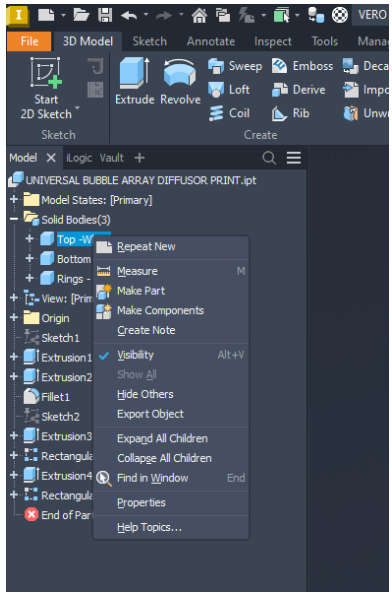


Figure 6.

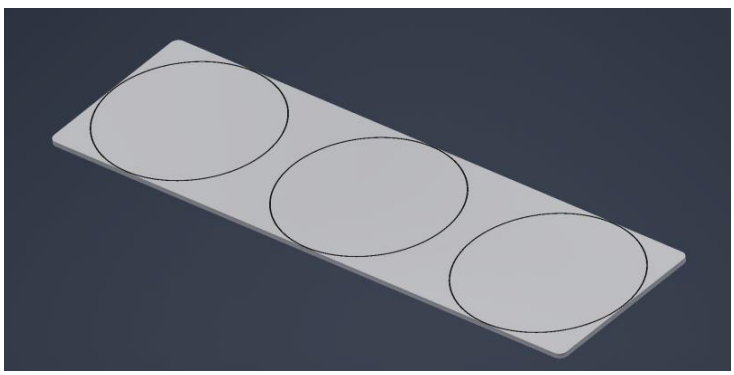


Figure 7.

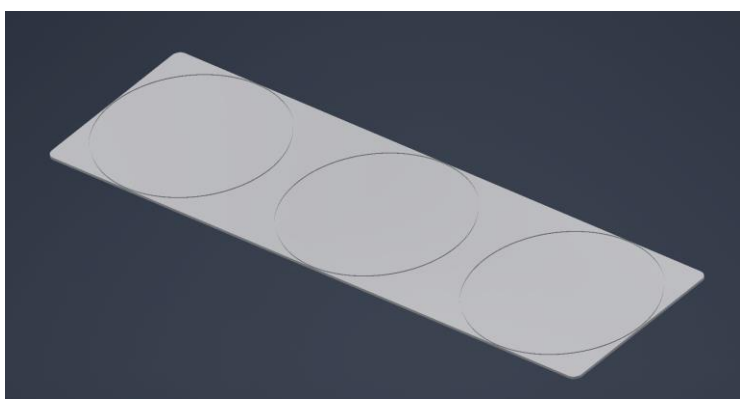


Figure 8.

Once only one solid body is visible, it can be exported as an .stl by navigating to the File tab on the main menu ribbon, clicking Export, and clicking CAD Format (Fig. 9). A Save As window will now pop-up. Use the drop down menu to select .stl as the save as type (Fig. 10). The name of the inventor .ipt part will auto-populate as the suggested name for the save as file. Be sure to add something that specifies which component of the part this file is, because multiple .stl files will be exported for each part and they must all have unique names. Example names for the three components of this file are shown in Figure 11. All remaining sold bodies can be exported using this same procedure – right clicking on the body of interest, selecting "hide others," and then exporting as an .stl.

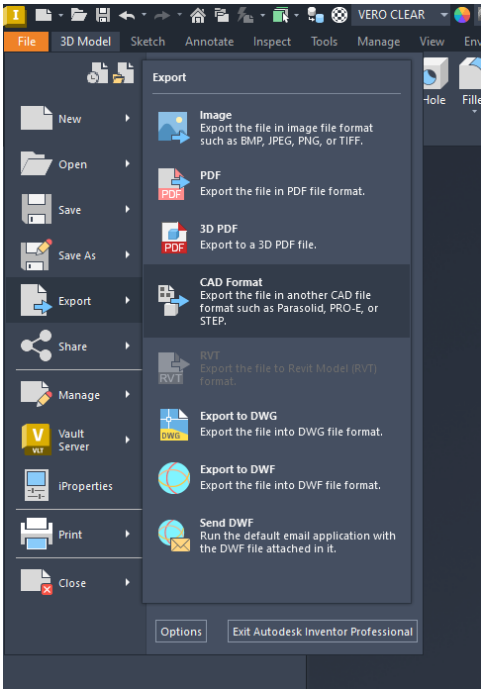


Figure 9.

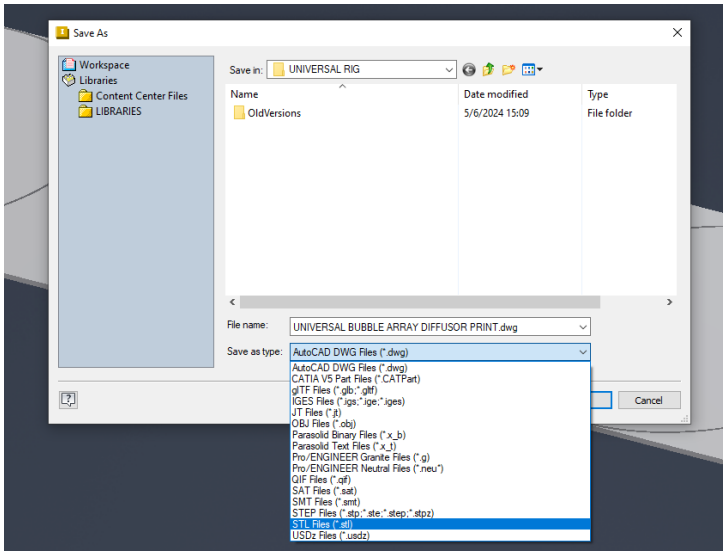
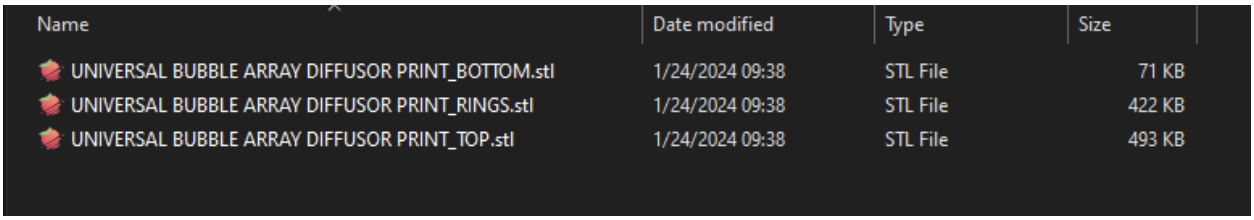


Figure 10.



Name	Date modified	Type	Size
UNIVERSAL BUBBLE ARRAY DIFFUSOR PRINT_BOTTOM.stl	1/24/2024 09:38	STL File	71 KB
UNIVERSAL BUBBLE ARRAY DIFFUSOR PRINT_RINGS.stl	1/24/2024 09:38	STL File	422 KB
UNIVERSAL BUBBLE ARRAY DIFFUSOR PRINT_TOP.stl	1/24/2024 09:38	STL File	493 KB

Figure 11.

GrabCAD Print is the software used to print parts on the J850 printer. With the software open, either click “ctrl + shift + A” or navigate to the File tab of the top menu ribbon and click “Add As Assembly” (Fig. 12). A file finder window will pop-up. Navigate to the folder where all the .stl files are stored. Hold the “ctrl” key while clicking each of the files that should be opened together or click and drag to highlight all the files (Fig. 13). Once the files are selected, press the “enter” key or the Open button in the bottom right corner of the screen. The files should now be open in the viewer and listed as an assembly in the New Project browser on the lefthand side of the screen (Fig. 14).

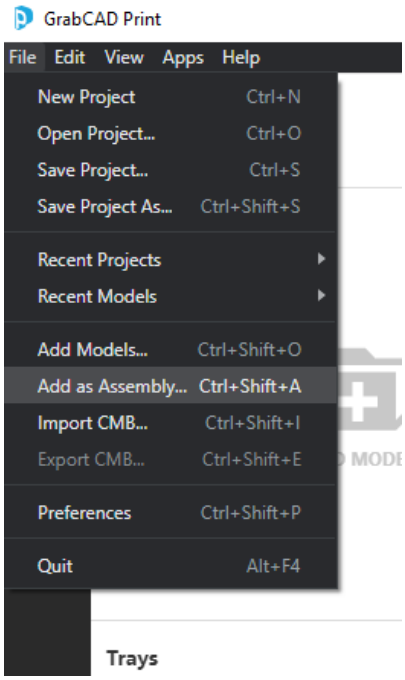


Figure 12.

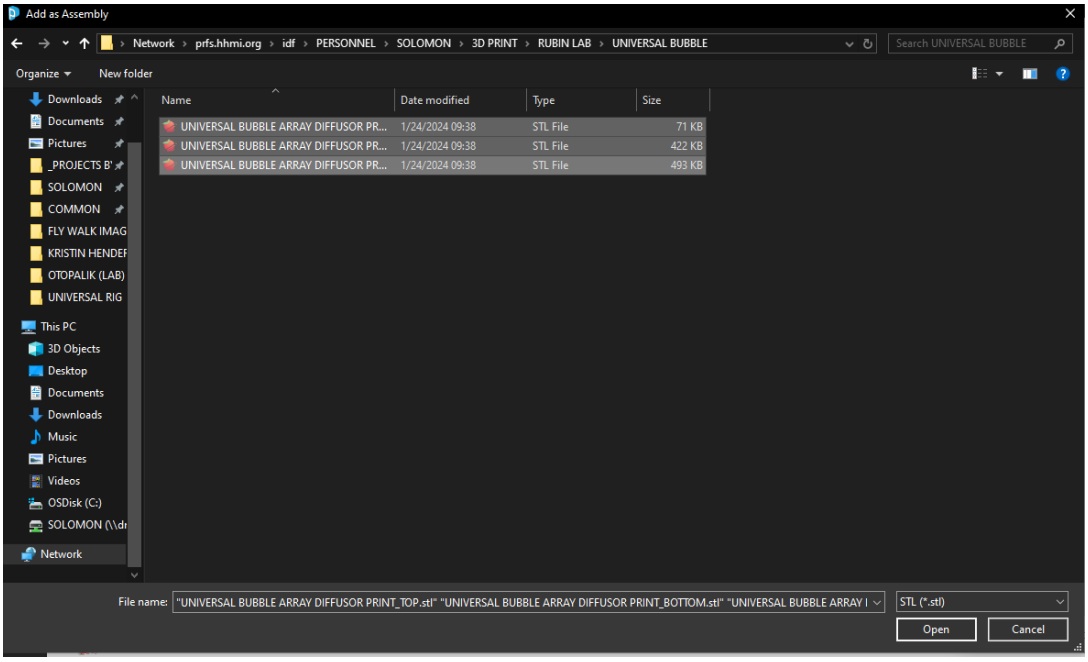


Figure 13.

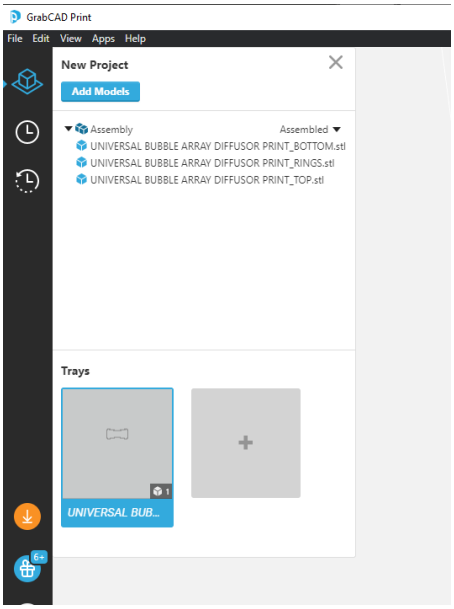


Figure 14.

When the files first open in the software, they will be way too big to fit on the print bed (Fig. 15). This can be resolved by changing the units from inches to mm. Click the Model Info icon on the righthand toolbar (Fig. 16) and select mm from the Units dropdown menu. The model should now fit easily within the build volume of the printer (Fig. 17).

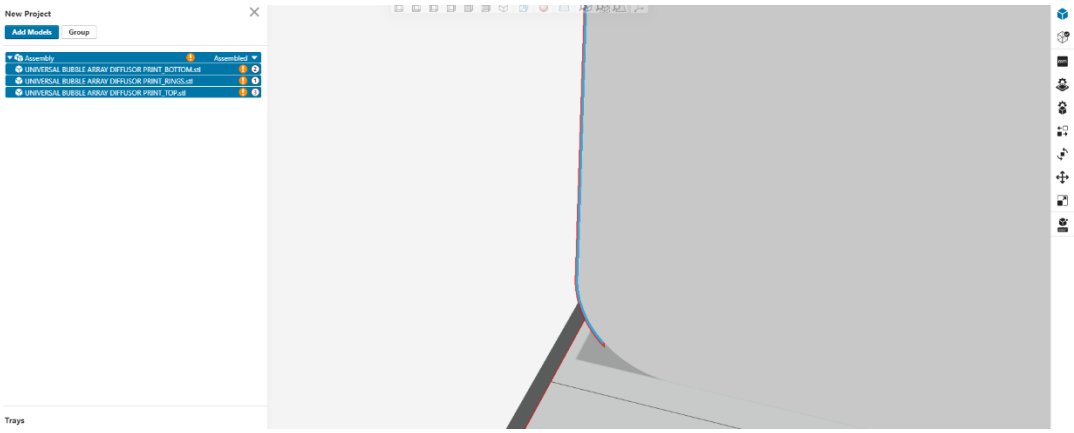


Figure 15.

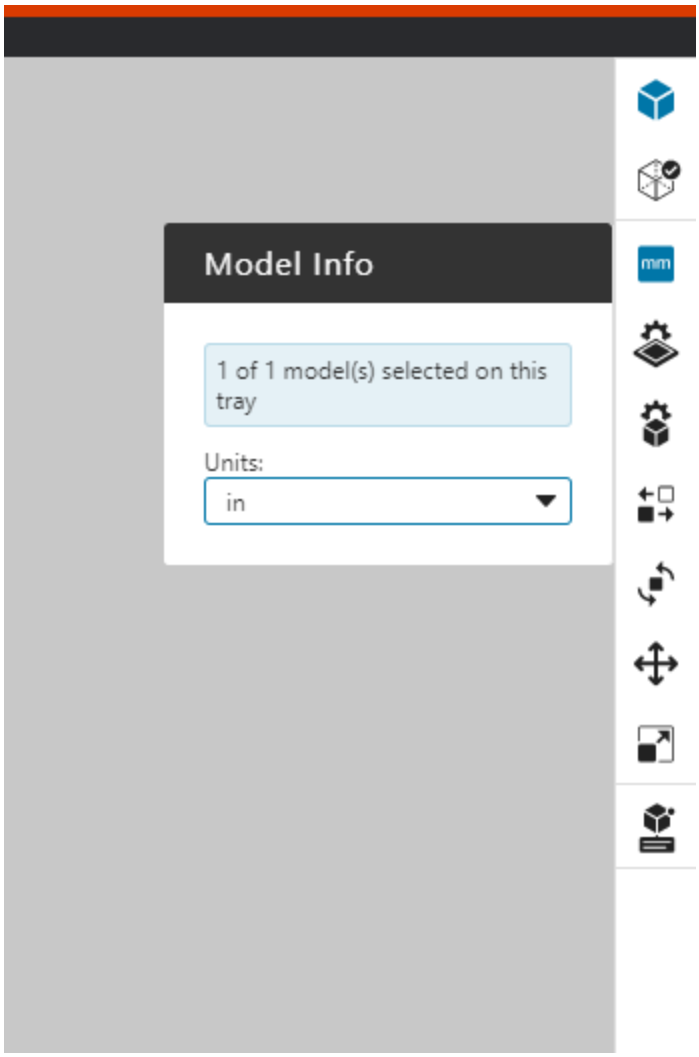


Figure 16.

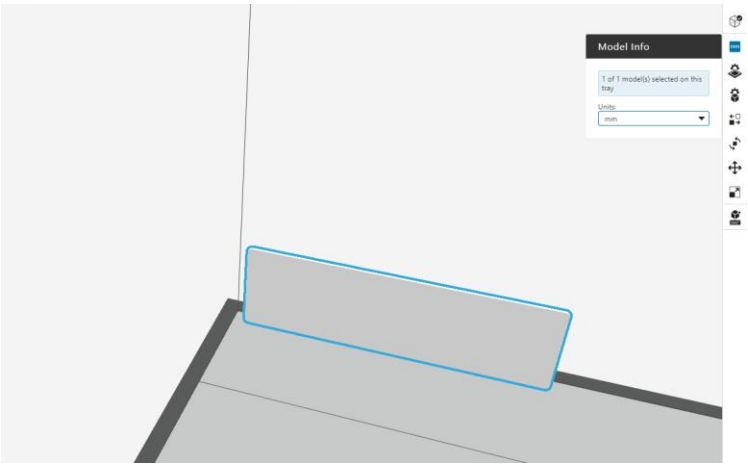


Figure 17.

Once the model is appropriately sized, the orientation can be adjusted. With the entire assembly selected, click the Orient icon on the righthand toolbar and then click the Orient Face to Plane Button (Fig. 18). Click the long, flat side of the bottom .stl and select Bottom in the Orient to Plane window (Fig. 19). The model orientation should now match Figure 20.

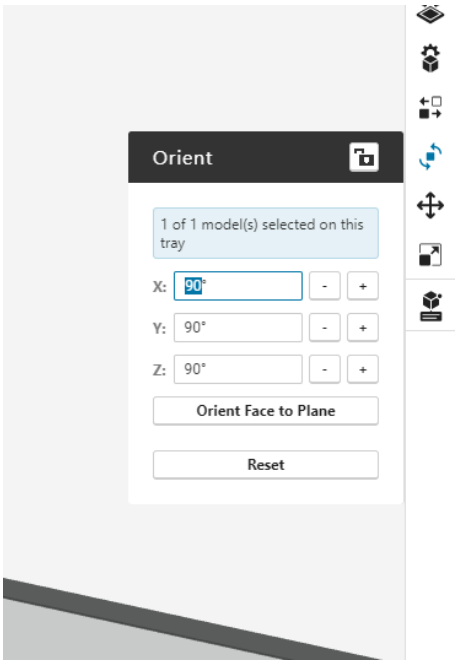


Figure 18.

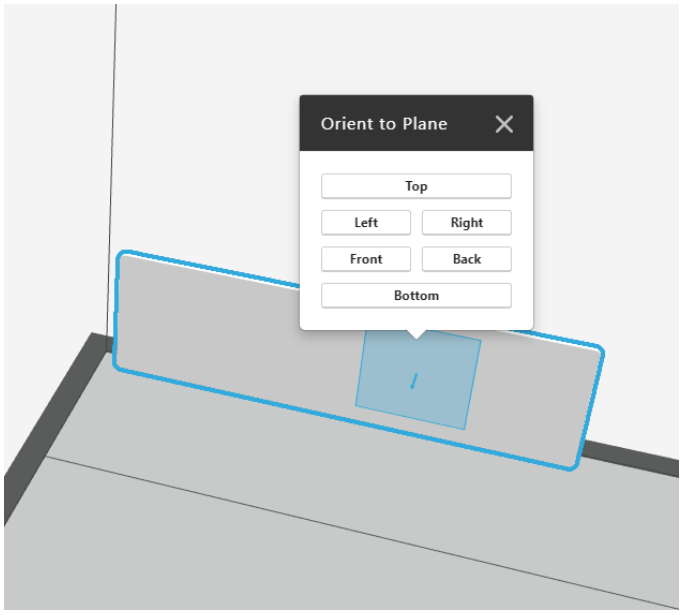


Figure 19.

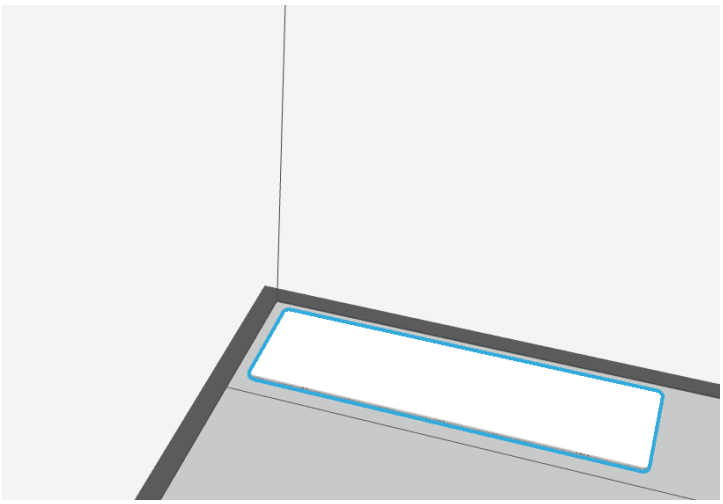


Figure 20.

The material for each individual .stl can be set using the Model Settings menu. With a single .stl file highlighted in the lefthand New Project browser, click the Model Settings icon on the righthand toolbar. The first dropdown menu should read Tray Materials, if it does not, select that option. The materials currently loaded in the printer will be listed below the dropdown as colored squares (Fig. 21). Hover over a square and a window will pop-up which lists the material type. Assign each .stl a material type by clicking on the desired material square in the Model Settings browser of the righthand toolbar while the .stl in question is highlighted in the New Project browser on the left (Fig. 22). Once the materials have all been assigned, confirm that the Finish type for all components is Matte. This settings is found in the bottom drop down menu of this Model Settings page. The job is now ready to be printed. Send the job to the printer by clicking the big, blue Print button in the bottom righthand corner of the screen (Fig. 23).

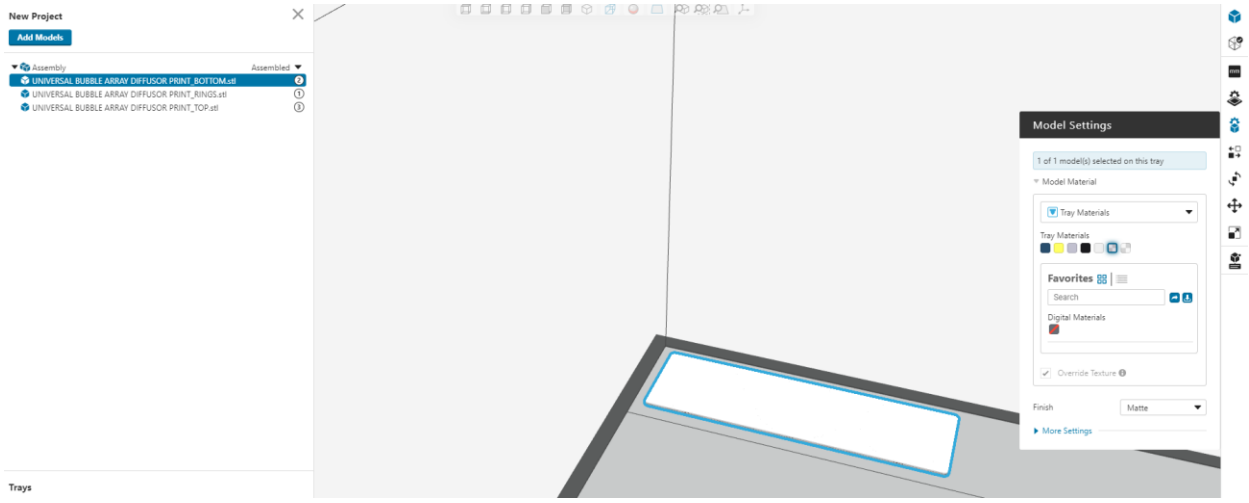


Figure 21.

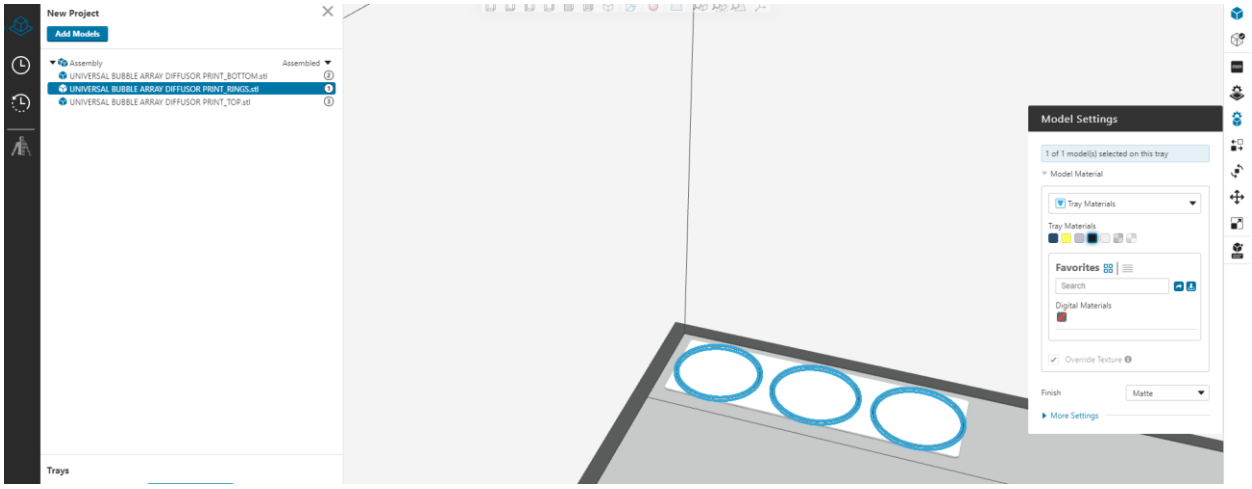


Figure 22.

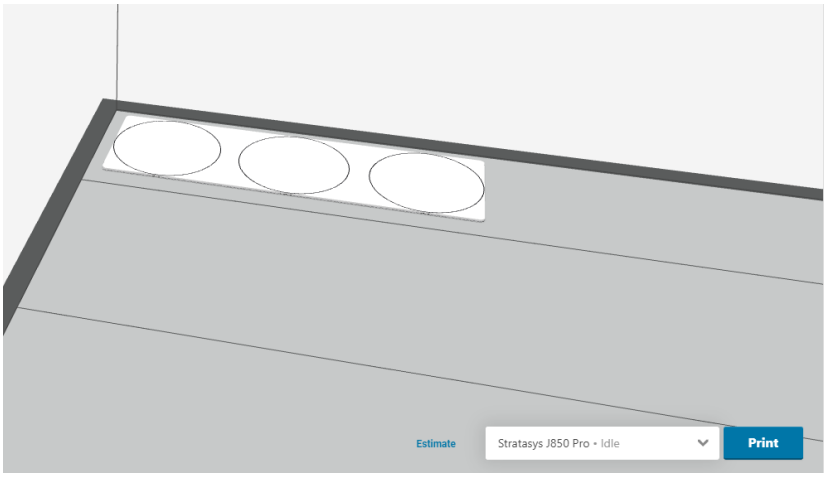


Figure 23.

Matte printing is achieved by encasing the printed object in support material. After the parts have been printed, the support needs to be removed. The Universal Bubble Array Diffusor Print can be carefully cleaned using the Stratasys pressure washer by fully supporting the back of the part with one hand and holding the water nozzle at a non-90-degree angle to the part, so that the water is running across the surface of the part rather than perpendicular to it. It is not recommended to clean the 30 and 90mm bowls in the pressure washer – they are easily punctured. In lieu of the pressure washer, most of the support can be removed by hand. The support is water soluble. Soaking the parts in a bowl of warm water for at least 10 minutes will make the support easier to remove. Once the majority of the support has been removed, re-soak the parts in a warm water or the lye bath for 10 minutes. Then, use a medium-stiffness bristle brush or your fingernails to scrape off any remaining support.